Koa Shen

Mechanical Engineer
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EDUCATION

University of California, Santa Barbara — Expected June 2028
B.S. Mechanical Engineering | Major GPA: 3.97 | Cumulative GPA: 3.94 — Santa Barbara, CA

- Dean's Honors every quarter. Coursework: Mechanics of Materials, Statics, Dynamics, Circuits, MATLAB.

EXPERIENCE

Fabrication Team Lead — Fall 2024 - Present
Gaucho Racing, Formula SAE Electric | UC Santa Barbara — Santa Barbara, CA

- CNC-machined 33 GR26 suspension components on Haas TM mills and lathes using Mastercam: 8 clevises and 16 rod ends from 4130 steel, plus 5 spindles and 4 uprights from 7075-T6 aluminum.
- Held clevis and rod-end bores within a 0.0004-in total tolerance window at about five-minute cycle times, and spindle and upright bearing interfaces within 0.0005 in; verified critical fits with go/no-go tests.
- Designed custom softjaws with two locating faces to hold repeatable datums across spindle operations; fixtured remaining operations with breadboard step-clamp setups.
- Fed GD&T and DFM constraints into suspension design reviews; standardized all five spindles on one tripod-coupling housing so proven CAM programs could be reused, likely preventing at least two more scrapped parts.
- Led 15 teammates through GR26 driver-safety-system design, fabrication, and installation, including the firewall, floor closeout, heat insulation, and composite seat; the car passed FSAE technical inspection.

Undergraduate Researcher — Fall 2025 - Present
Hawkes Lab, UC Santa Barbara — Santa Barbara, CA

- Designed and 3D-printed custom molds with complex curvature for composite parts; prepared molds and fabricated parts by wet layup and prepreg across different weaves and fabrics.
- Experimented with vacuum-bagging setups and iterated on mold design to improve part stiffness.

Maintenance Technician — Summer 2025
Exploratorium — San Francisco, CA

- Serviced 100+ public exhibits and led three major overhauls, reverse-engineering failing components and fabricating replacements with manual machining, welding, and woodshop equipment.
- Reverse-engineered lost viewer CAD from an existing unit, applied GD&T, and machined replacement batches on a Haas CNC router with CAM and fixturing chosen to reduce material waste.
- Redesigned sand-damaged encoder mounts with a thicker flange and toleranced outer diameter, then manually machined them onsite using a clocking setup for blind tapped holes.
- Replaced over-constraining fasteners with a load-spreading bezel to stop acrylic enclosure cracking, avoiding $3,000+ in replacement costs; left CAD, CAM files, and remanufacturing instructions in Confluence.

Mechanical Engineering Intern — Summer 2026 - Present
SLAC National Accelerator Laboratory, LCLS — Menlo Park, CA (part-time remote)

- Released Siemens Solid Edge designs and Teamcenter PDM-controlled drawings with GD&T for manufacture and inspection of beamline motion assemblies.
- Redesigned a high-use optics assembly with new brackets, kinematic mounts, and detector strain relief, guided by a Python/Drake collision simulation that exposed at least five physical pinch points.

PROJECTS

FSAE Chassis Welding Fixtures | Gaucho Racing — Winter 2024 - Spring 2026

- Led a three-person team to build the GR25 fixture on a $1,500 budget; 80/20 and laser-cut plywood located 81 chassis tubes within 0.050 in of CAD nodes and cut assembly time 50%.
- Redesigned the GR26 fixture around 3D-printed locating brackets and reused 80/20, costing less than laser-cut equivalents and cutting assembly time a further 30%.

TECHNICAL SKILLS

Manufacturing: Haas CNC mills and lathes, Mastercam, CAM, manual machining, CNC router, fixturing, softjaws, welding, tube notching, sheet metal, composite layup, mold making, 3D printing
Quality and Design: GD&T, DFM, go/no-go inspection, tolerance stackup, SolidWorks, Solid Edge, Teamcenter PDM, FEA, engineering drawings
Software: Python, C/C++, MATLAB, Drake, Git